Network Layer Protocols and IP Addressing

Foreword

- Internet Protocol Version 4 (IPv4) is the core protocol suite in the TCP/IP protocol suite. It works at the network layer in the TCP/IP protocol stack and this layer corresponds to the network layer in the Open System Interconnection Reference Model (OSI RM).
- The network layer provides connectionless data transmission services. A network does not need to establish a connection before sending data packets. Each IP data packet is sent separately.
- This presentation describes the basic concepts of IPv4 addresses, subnetting, network IP address planning, and basic IP address configuration.

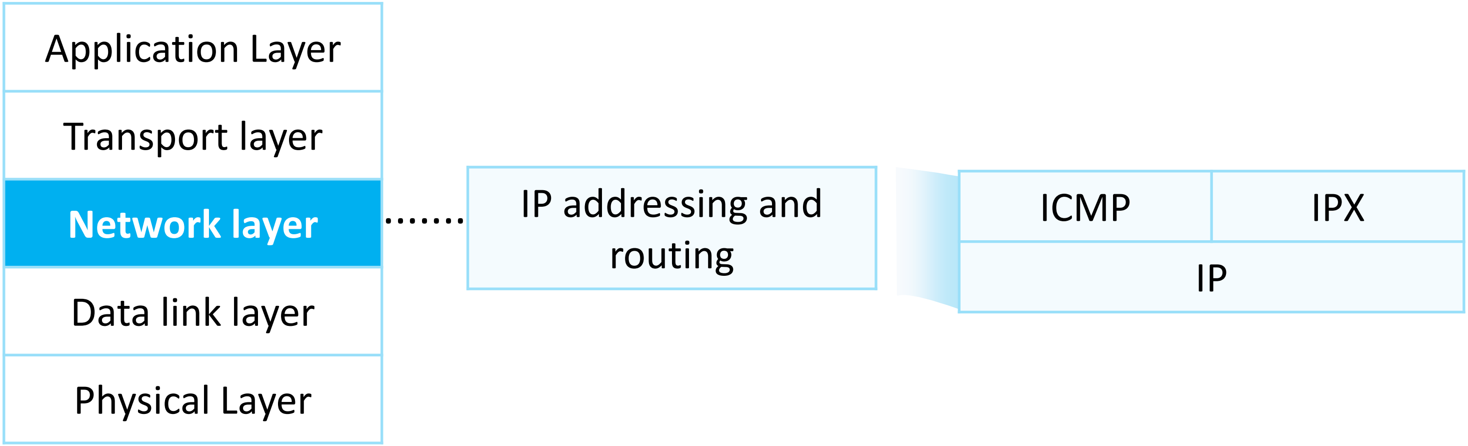
Objectives

- On completion of this course, you will be able:
  - Describe main protocols at the network layer.
  - Describe the concepts and classification of IPv4 addresses and special IPv4 addresses.
  - Calculate IP networks and subnets.
  - Use the IP network address planning method.

Contents

Network Layer Protocols

- The network layer is often called the IP layer. Network layer protocols include Internet Control Message Protocol (ICMP) and Internet Packet Exchange (IPX), in addition to IP.



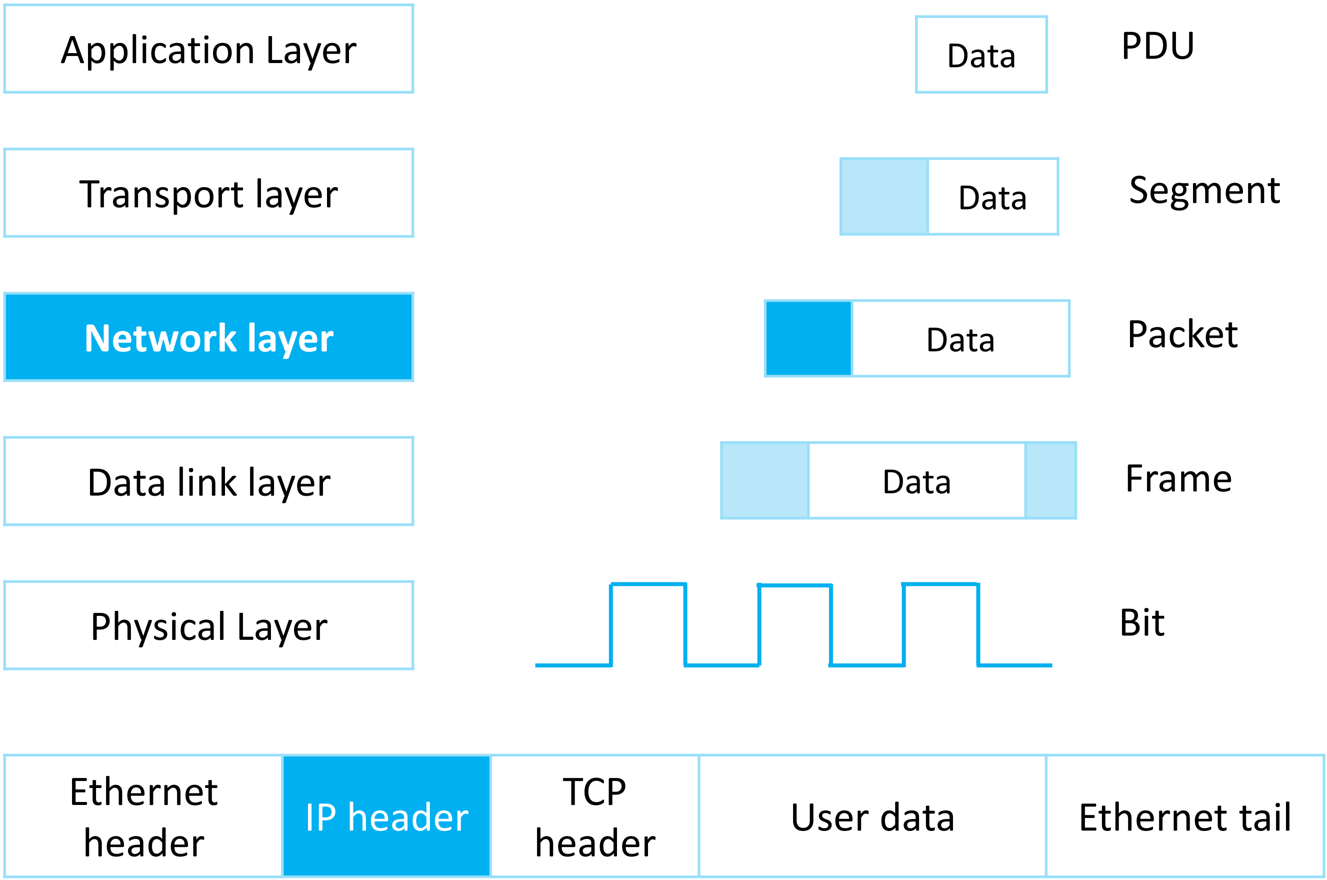
Equivalent TCP/IP model

Internet Protocol

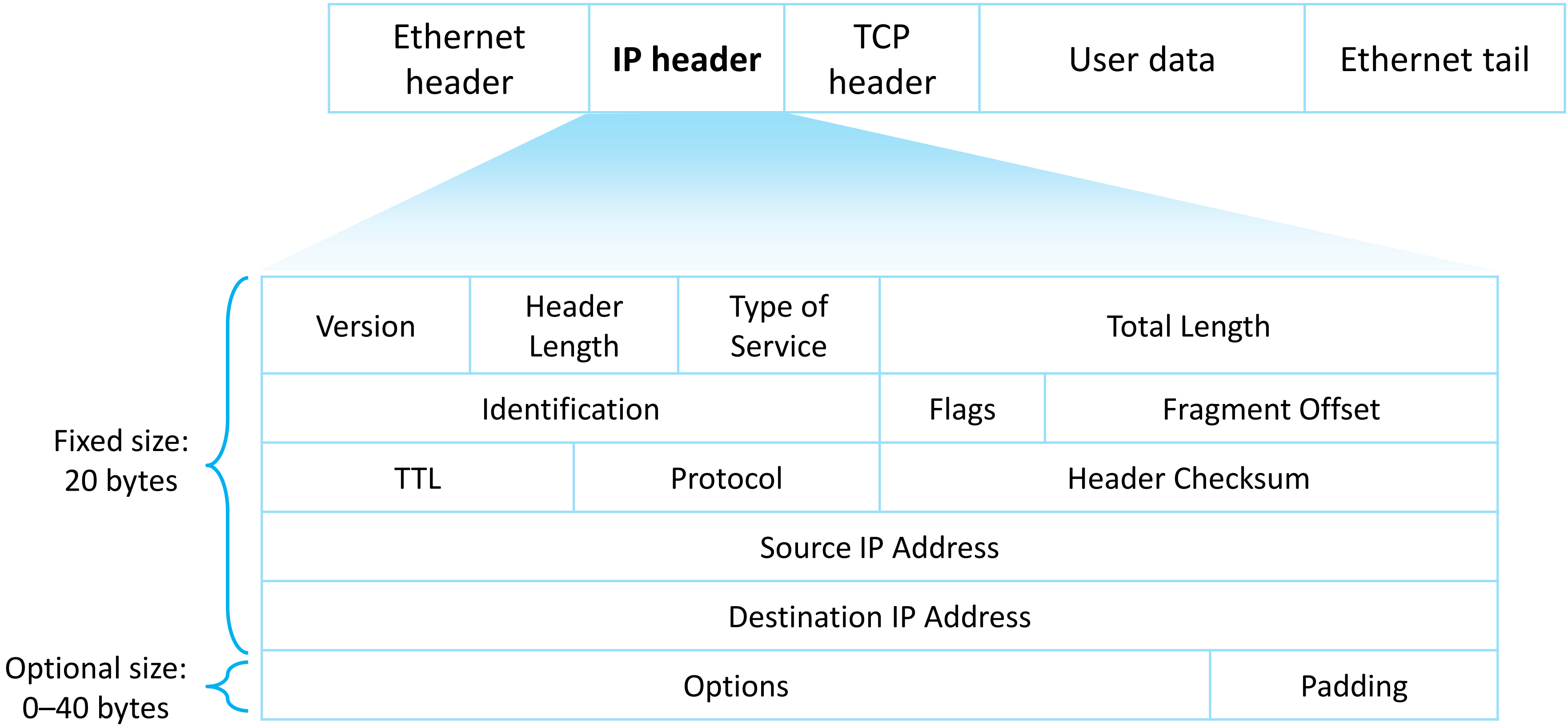
- IP is short for the Internet Protocol. IP is the name of a protocol file with small content. It defines and describes the format of IP packets.
- The frequently mentioned IP refers to any content related directly or indirectly to the Internet Protocol, instead of the Internet Protocol itself.

Function	Version
• Provides logical addresses for devices at the network layer. • Is responsible for addressing and forwarding data packets.	• IP Version 4 (IPv4) • IP Version 6 (IPv6)

Data Encapsulation



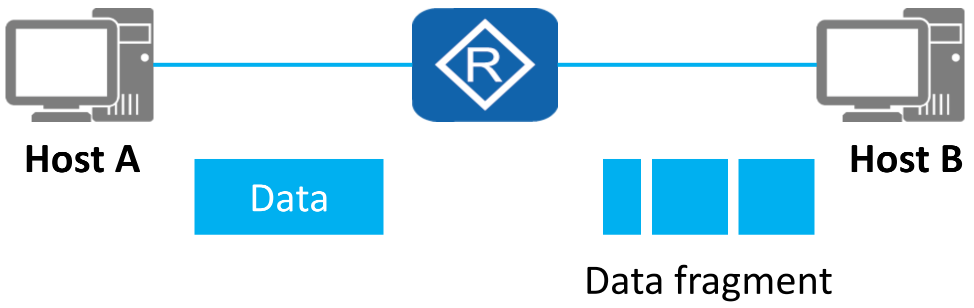
IPv4 Packet Format



Data Packet Fragmentation

- The process of dividing a packet into multiple fragments is called fragmentation.
- The sizes of IP packets forwarded on a network may be different. If the size of an IP packet exceeds the maximum size supported by a data link, the packet needs to be divided into several smaller fragments before being transmitted on the link.

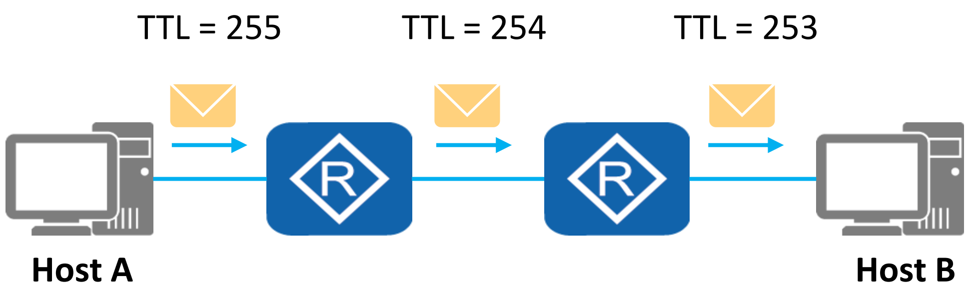
<table>
<tr><td>Version</td><td>Header Length</td><td>Type of Service</td><td colspan="2">Total Length</td></tr>
<tr><td colspan="3">Identification</td><td>Flags</td><td>Fragment Offset</td></tr>
<tr><td>TTL</td><td colspan="2">Protocol</td><td colspan="2">Header Checksum</td></tr>
<tr><td colspan="5">Source IP Address</td></tr>
<tr><td colspan="5">Destination IP Address</td></tr>
<tr><td colspan="4">Options</td><td>Padding</td></tr>
</table>



Time to Live

- The TTL field specifies the number of routers that a packet can pass through.
- Once a packet passes through a router, the TTL is reduced by 1. If the TTL value is reduced to 0, a data packet is discarded.

Version	Header Length	Type of Service	Total Length	
Identification			Flags	Fragment Offset
TTL	Protocol		Header Checksum	
Source IP Address				
Destination IP Address				
Options				Padding



Protocol

- The Protocol field in the IP packet header identifies a protocol that will continue to process the packet.
- This field identifies the protocol used by the data carried in the data packet so that the IP layer of the destination host sends the data to the process mapped to the Protocol field.

<table>
<tr><td>Version</td><td>Header Length</td><td>Type of Service</td><td colspan="3">Total Length</td></tr>
<tr><td colspan="3">Identification</td><td>Flags</td><td colspan="2">Fragment Offset</td></tr>
<tr><td colspan="2">TTL</td><td>Protocol</td><td colspan="3">Header Checksum</td></tr>
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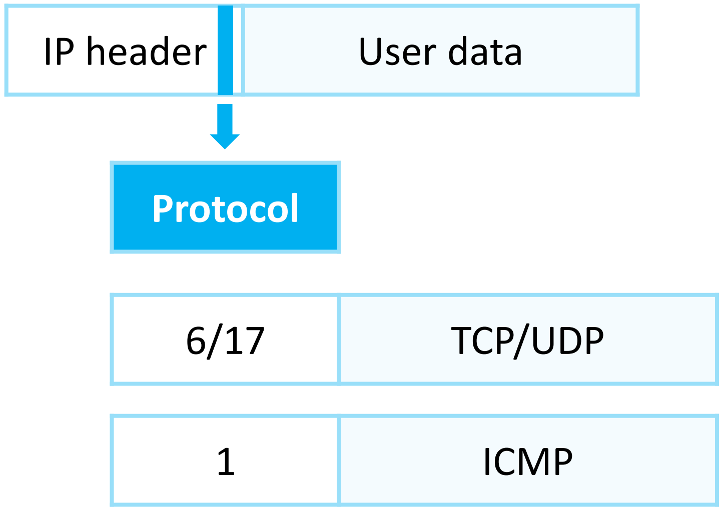

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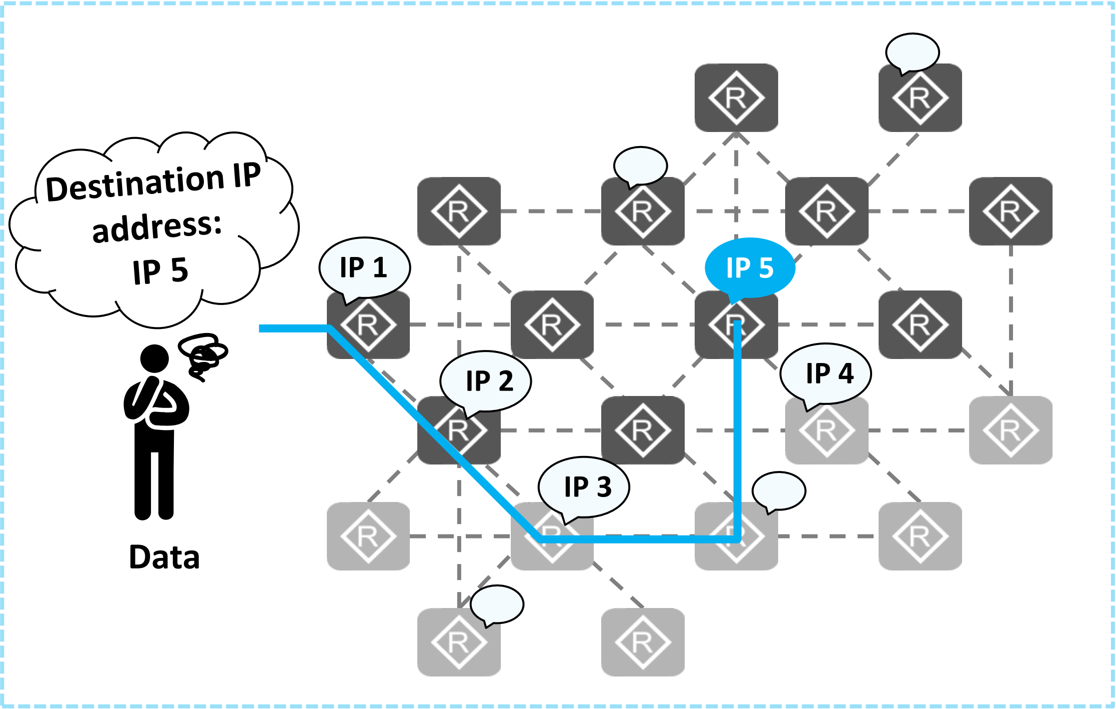
Contents

What Is an IP Address?

- An IP address identifies a node (or an interface on a network device) on a network.
- IP addresses are used to forward IP packets on the network.



IP Address

An IP address identifies a node on a network and is used to find the destination for data.

IP address Notation

- An IPv4 address is 32 bits long.
- It is in dotted decimal notation.

Dotted decimal notation

Decimal	**192.**	**168.**	**10.**	**1**	4 bytes
Binary	**11000000**	10101000	00001010	00000001	32 bits

Conversion between decimal and binary systems

Power	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
	128	64	32	16	8	4	2	1
Bit	1	1	0	0	0	0	0	0

= 128 + 64 = 192

- IPv4 address range is 0.0.0.0–255.255.255.255.

IP Address Structure

- **Network part:** identifies a network.
- **Host part:** identifies a host and is used to differentiate hosts on a network.

Network part	Host part

- **Network mask:** is used to distinguish the network part from the host part in an IP address.

192.	168.	10.	1

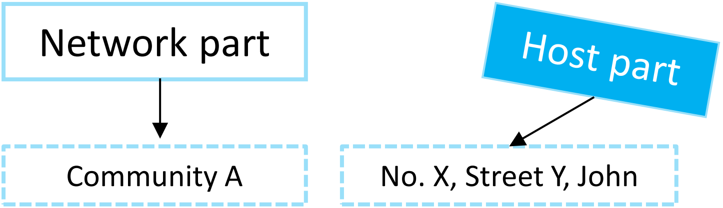
192.168.10.1	1 1 0 0 0 0 0 0	1 0 1 0 1 0 0 0	0 0 0 0 1 0 1 0	0 0 0 0 0 0 0 1	**IP address**
255.255.255.0	1 1 1 1 1 1 1 1	1 1 1 1 1 1 1 1	1 1 1 1 1 1 1 1	0 0 0 0 0 0 0 0	**Network mask**
	Network part			Host part	

Written as

192.168.10.1 255.255.255.0 = 192.168.10.1/24

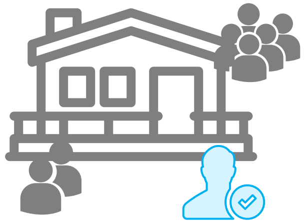
IP Addressing

- **Network part (network ID):** identifies a network.
- **Host part:** identifies a host and is used to differentiate hosts on a network.



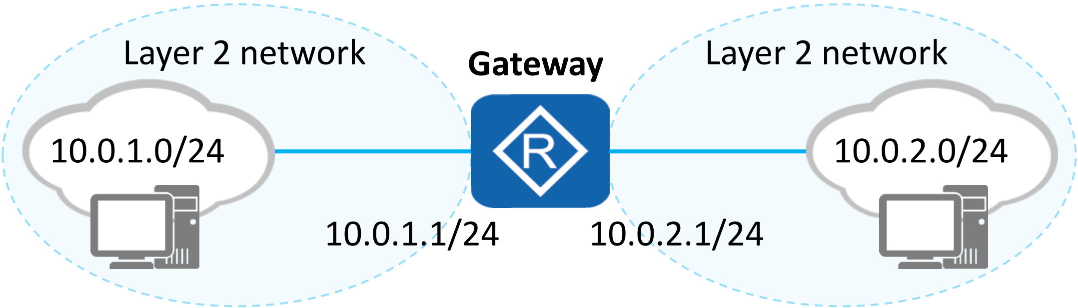
Layer 2 network addressing

Community A (network bits)



No. X, Street Y, John
(host bits)

Layer 3 network addressing



Layer 3 network

IP Address Classification (Classful Addressing)

- To facilitate IP address management and networking, IP addresses are classified into the following classes:

Class A	0NNNNNNN	NNNNNNNN	NNNNNNNN	NNNNNNNN	0.0.0.0–127.255.255.255	Assigned to hosts
Class B	10NNNNNN	NNNNNNNN	NNNNNNNN	NNNNNNNN	128.0.0.0–191.255.255.255	
Class C	110NNNNN	NNNNNNNN	NNNNNNNN	NNNNNNNN	192.0.0.0–223.255.255.255	
Class D	**1110**NNNN	NNNNNNNN	NNNNNNNN	NNNNNNNN	224.0.0.0–239.255.255.255	Used for multicast
Class E	**1111**NNNN	NNNNNNNN	NNNNNNNN	NNNNNNNN	240.0.0.0–255.255.255.255	Used for research

- Default subnet masks of classes A, B, and C
  - Class A: 8 bits, 0.0.0.0–127.255.255.255/8
  - Class B: 16 bits, 128.0.0.0–191.255.255.255/16
  - Class C: 24 bits, 192.0.0.0-223.255.255.255/24

Network part

Host part

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IP Address Types

- A network range defined by a network ID is called a network segment.
- **Network address**: identifies a network.

  Example: 192.168.10.0/24

192.	168.	10.	00000000

- **Broadcast address**: a special address used to send data to all hosts on a network.

  Example: 192.168.10.255/24

192.	168.	10.	11111111

- **Available addresses**: IP addresses that can be allocated to device interfaces on a network.

  Example: 192.168.10.1/24

192.	168.	10.	00000001

Note

- Network and broadcast addresses cannot be directly used by devices or their interfaces.
- Number of available addresses on a network segment is $2^n - 2$ (n is the number of bits in the host part).

IP Address Calculation

- Example: What are the network address, broadcast address, and number of available addresses of class B address 172.16.10.1/16?

172.	16.	00001010.	00000001

	Octet 1	Octet 2	Octet 3	Octet 4	
IP address	1 0 1 0 1 1 0 0	0 0 0 1 0 0 0 0	0 0 0 0 1 0 1 0	0 0 0 0 0 0 0 1	
Network mask	1 1 1 1 1 1 1 1	1 1 1 1 1 1 1 1	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0	
Network address	1 0 1 0 1 1 0 0	0 0 0 1 0 0 0 0	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0	The network address is obtained, with all host bits set to 0s. 172.16.0.0/16
Broadcast address	1 0 1 0 1 1 0 0	0 0 0 1 0 0 0 0	1 1 1 1 1 1 1 1	1 1 1 1 1 1 1 1	The broadcast address is obtained, with all host bits set to 1s. 172.16.255.255/16

Number of IP addresses $2^{16} = 65536$

Number of available addresses $2^{16} - 2 = 65534$

Range of available addresses 172.16.0.1/16–172.16.255.254/16

Quiz

Example: What are the network address, broadcast address, and number of available addresses of class A address 10.128.20.10/8?

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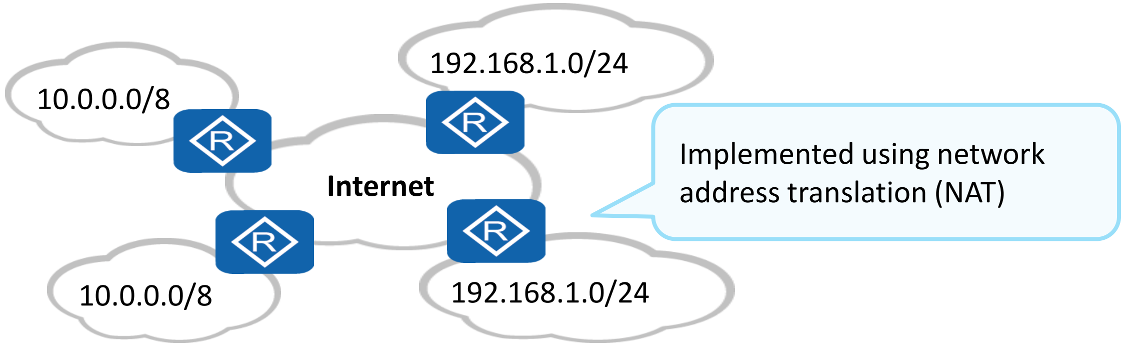
Private IP Addresses

- **Public IP address:** An IP address is assigned by the Internet Assigned Numbers Authority (IANA), and this address allocation mode ensures that each IP address is unique on the Internet. Such an IP address is a public IP address.
- **Private IP address:** In practice, some networks do not need to connect to the Internet. For example, on a network of a lab in a college, IP addresses of devices need to avoid conflicting with each other only within the same network. In the IP address space, some IP addresses of class A, B, and C addresses are reserved for the preceding situations. These IP addresses are called private IP addresses.
  - Class A: 10.0.0.0–10.255.255.255
  - Class B: 172.16.0.0–172.31.255.255
  - Class C: 192.168.0.0–192.168.255.255



Connecting a private network to the Internet

Special IP Addresses

- Some IP addresses in the IP address space are of special meanings and functions.
- For example:

Special IP Address	Address Scope	Function
Limited broadcast address	255.255.255.255	It can be used as a destination address and traffic destined for it is sent to all hosts on the network segment to which the address belongs. (Its usage is restricted by a gateway).
Any IP address	0.0.0.0	It is an address of any network. Addresses in this block refer to source hosts on "this" network.
Loopback address	127.0.0.0/8	It is used to test the software system of a test device.
Link-local address	169.254.0.0/24	If a host fails to automatically obtain an IP address, the host can use an IP address in this address block for temporary communication.

IPv4 vs. IPv6

- IPv4 addresses managed by the IANA were exhausted in 2011. As the last public IPv4 address was allocated and more and more users and devices access the public network, IPv4 addresses were exhausted. This is the biggest driving force for IPv6 to replace IPv4.

IPv4	IPv6
• Address length: 32 bits • Address types: unicast address, broadcast address, and multicast address • Characteristics: ▫ IPv4 address depletion ▫ Inappropriate packet header design ▫ ARP dependency-induced flooding ▫ ...	• Address length: 128 bits • Address types: unicast address, multicast address, and anycast address • Characteristics: ▫ Unlimited number of addresses ▫ Simplified packet header ▫ Automatic IPv6 address allocation ▫ ...

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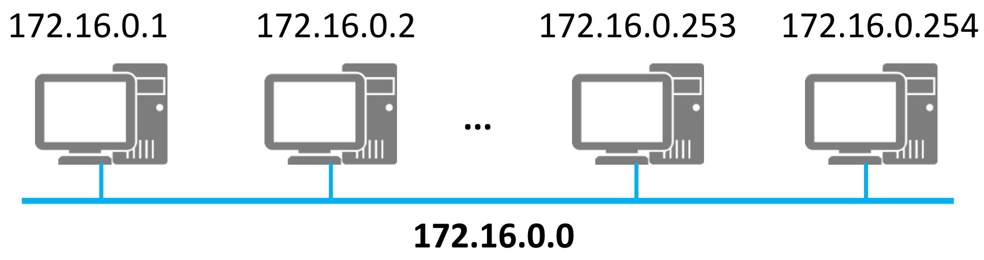
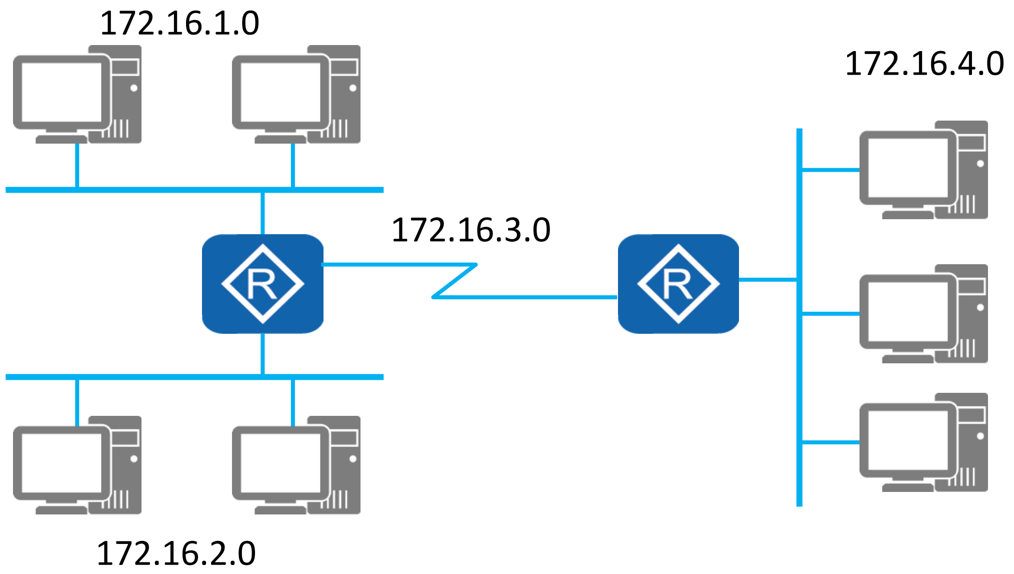
Why Subnetting?



2^{16} = 65536 IP addresses

- A class B address is used for a broadcast domain, wasting addresses.
- The broadcast domain is too large. Once broadcast occurs, an internal network is overloaded.



- A network number is divided into multiple subnets, and each subnet is allocated to a separate broadcast domain.
- In this way, the broadcast domain is smaller, and the network planning is more reasonable.
- IP addresses are properly used.

Subnetting - Analyzing the Original Network Segment

- Example: 192.168.10.0/24

192.168.10.1

IP address	192.	168.	10.	0	0	0	0	0	0	0	1
Default subnet mask	255.	255.	255.	0	0	0	0	0	0	0	0

...

192.168.10.255

IP address	192.	168.	10.	1	1	1	1	1	1	1	1
Default subnet mask	255.	255.	255.	0	0	0	0	0	0	0	0

Network part **Host part**

One class C network:

192.168.10.0/24

Default subnet mask:

255.255.255.0

Network address: 192.168.10.0/24

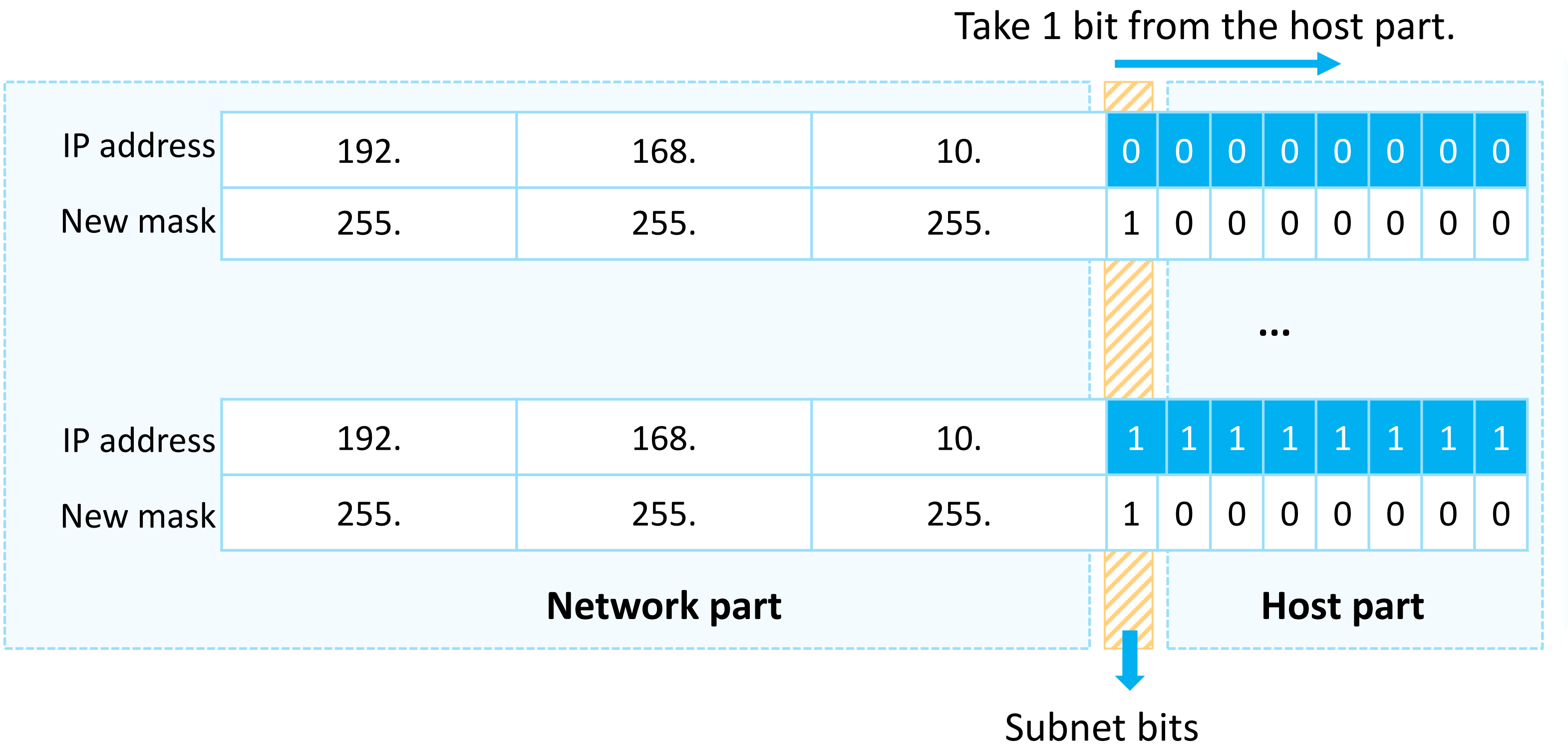
Broadcast address: 192.168.10.255

Total IP addresses: $2^8 = 256$

Available IP addresses: $2^8 – 2 = 254$

Subnetting - Taking Bits from the Host Part

- Bits can be taken from the host part to create subnets.



Two subnets:

Subnet 1: 192.168.10.0/25

Subnet 2: 192.168.10.128/25

New mask: 255.255.255.128

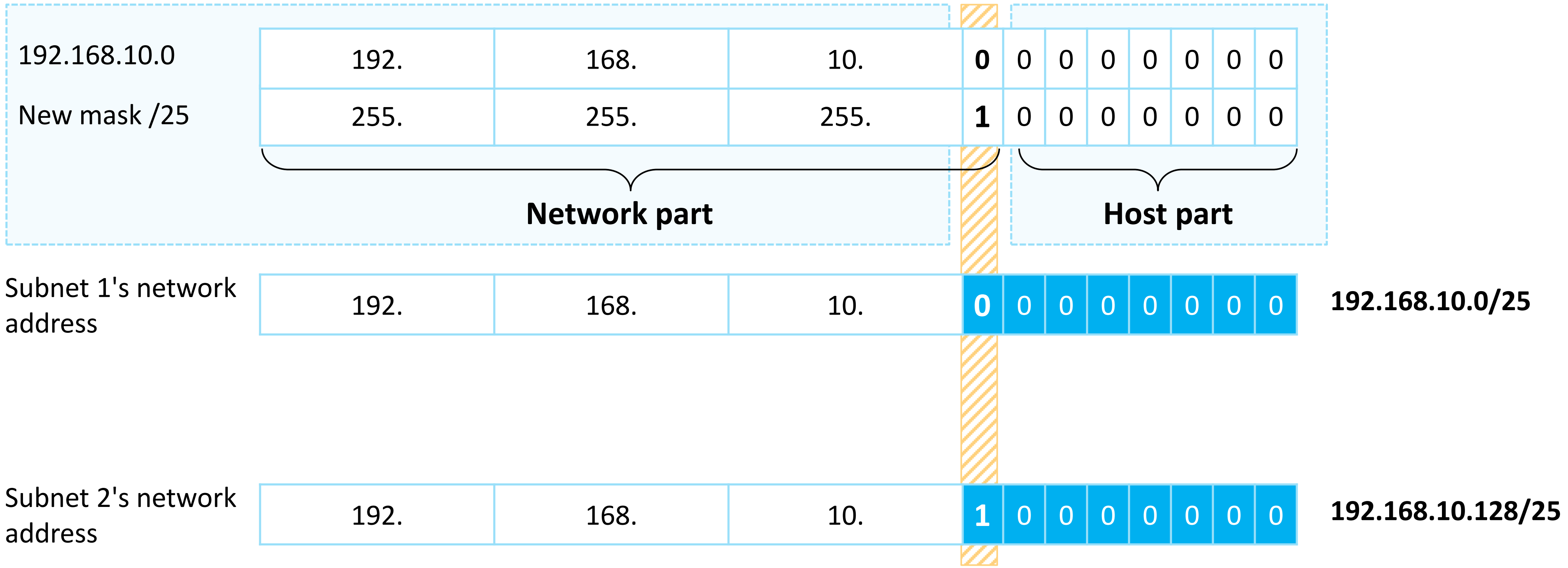
- Variable length subnet mask (VLSM)

Total IP addresses: $2^7 = 128$

Available IP addresses: $2^7 – 2 = 126$

Subnetting - Calculating the Subnet Network Address

- The network address is obtained, with all host bits set to 0s.



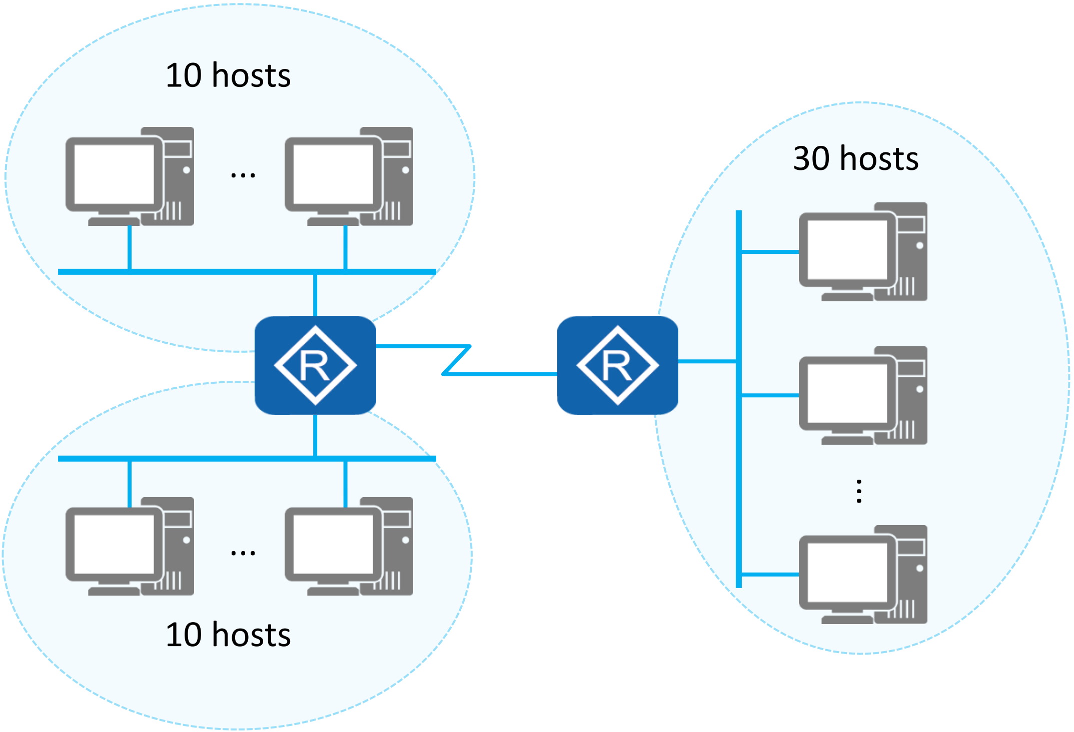
Subnetting - Calculating the Subnet Broadcast Address

- The broadcast address is obtained, with all host bits set to 1s.

192.168.10.0	192.	168.	10.	**0**	0	0	0	0	0	0	0
New mask /25	255.	255.	255.	**1**	0	0	0	0	0	0	0
	Network part				**Host part**						

Subnet 1's network address	192.	168.	10.	**0**	0	0	0	0	0	0	0	192.168.10.0/25
Subnet 1's broadcast address	192.	168.	10.	**0**	1	1	1	1	1	1	1	**192.168.10.127/25**
Subnet 2's network address	192.	168.	10.	**1**	0	0	0	0	0	0	0	192.168.10.128/25
Subnet 2's broadcast address	192.	168.	10.	**1**	1	1	1	1	1	1	1	**192.168.10.255/25**

Practice: Computing Subnets (1)



- **Question:** An existing class C network segment is 192.168.1.0/24. Use the VLSM to allocate IP addresses to three subnets.

- **Answer:** (Use a network with 10 hosts as an example.)

  Step 1: Calculate the number of host bits to be taken.

  $2^n - 2 \geq 10$

  $n \geq 4$, host bits

  Step 2: Take bits from the host part.

  Take 4 bits from the host part.

IP address	192.	168.	1.	**0**	**0**	**0**	**0**	0	0	0	0
Subnet mask	255.	255.	255.	1	1	1	1	0	0	0	0

Subnet bits

Number of subnets:
2^4 = 16 subnets

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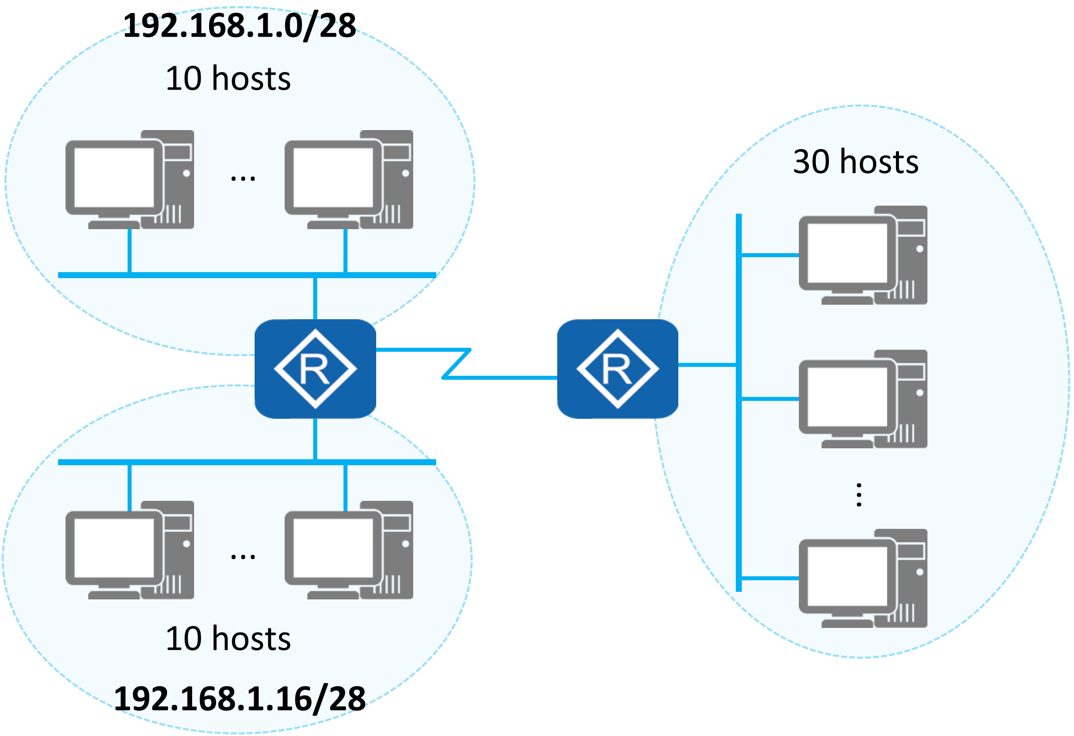
Practice: Computing Subnets (2)



- **Question:** An existing class C network segment is 192.168.1.0/24. Use the VLSM to allocate IP addresses to three subnets.

- **Answer:** (Use a network with 10 hosts as an example.)

  Step 3: Calculate subnet network addresses.

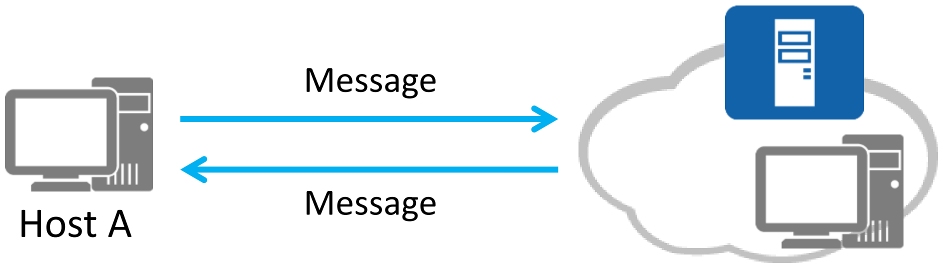
												Network address
IP address	192.	168.	1.	**0**	**0**	**0**	**0**	0	0	0	0	
New mask	255.	255.	255.	1	1	1	1	0	0	0	0	
Subnet 1	192.	168.	1.	0	0	0	0	0	0	0	0	192.168.1.0/28
Subnet 2	192	168.	1.	0	0	0	1	0	0	0	0	192.168.1.16/28
Subnet 3	192.	168.	1.	0	0	1	0	0	0	0	0	192.168.1.32/28
				⋮								
Subnet 16	192.	168.	1.	1	1	1	1	0	0	0	0	192.168.1.240/28

Contents

ICMP

- The Internet Control Message Protocol (ICMP) is an auxiliary protocol of the IP protocol.

- ICMP is used to transmit error and control information between network devices. It plays an important role in collecting network information, diagnosing and rectifying network faults.



Ethernet header	IP header	**ICMP message**	Ethernet tail

Type	**Code**	Checksum
ICMP message content		

Type	Code	Description
0	0	Echo Reply
3	0	Network Unreachable
3	1	Host Unreachable
3	2	Protocol Unreachable
3	3	Port Unreachable
5	0	Redirect
8	0	Echo Request

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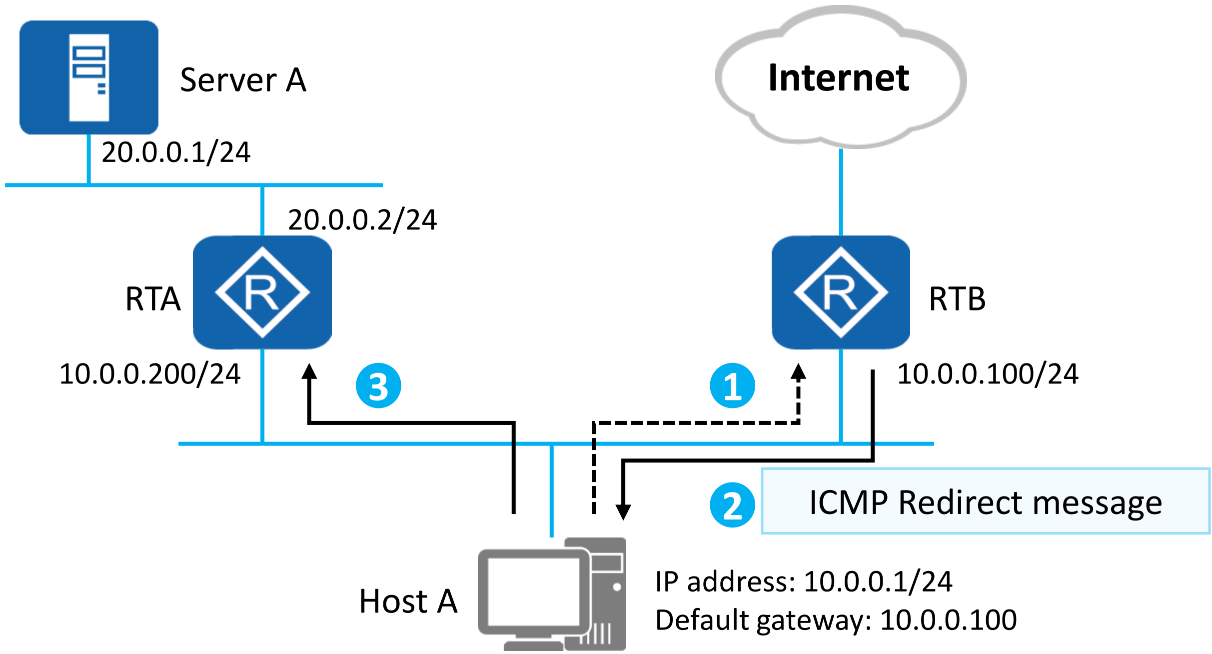
ICMP Redirection

- ICMP Redirect messages are a type of ICMP control message. When a router detects that a host uses a non-optimal route in a specific scenario, the router sends an ICMP Redirect message to the host, requesting the host to change the route.



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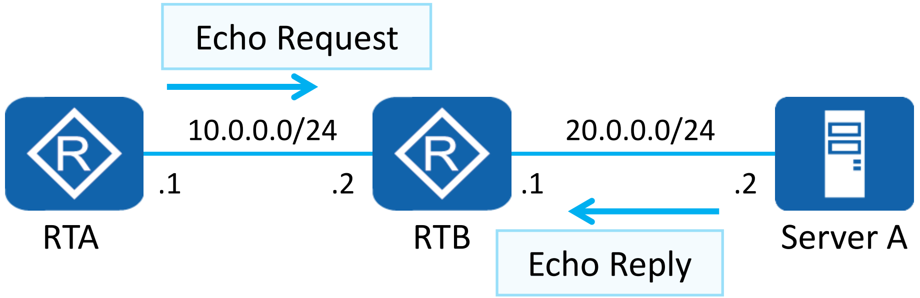
ICMP Error Detection

- ICMP Echo messages are used to check network connectivity between the source and destination and provide other information, such as the round-trip time.




```
[RTA]ping 20.0.0.2
 PING 20.0.0.2: 56  data bytes, press CTRL_C to break
  Reply from 20.0.0.2: bytes=56 Sequence=1 ttl=254 time=70 ms
  Reply from 20.0.0.2: bytes=56 Sequence=2 ttl=254 time=30 ms
  Reply from 20.0.0.2: bytes=56 Sequence=3 ttl=254 time=30 ms
  Reply from 20.0.0.2: bytes=56 Sequence=4 ttl=254 time=40 ms
  Reply from 20.0.0.2: bytes=56 Sequence=5 ttl=254 time=30 ms

 --- 20.0.0.2 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 30/40/70 ms
```

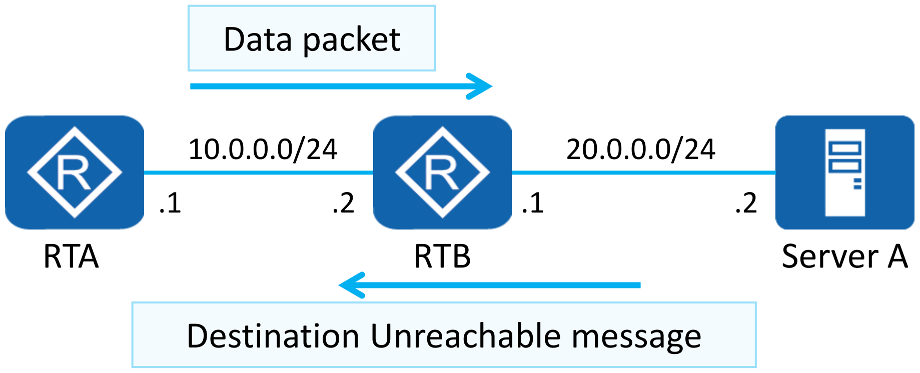
Function: Ping

Ping is a command used on network devices, Windows OS, Unix OS, and Linux OS. Ping is a small and useful application based on the ICMP protocol.

A ping tests the reachability of a destination node.

ICMP Error Report

- ICMP defines various error messages for diagnosing network connectivity problems. The source can determine the cause for a data transmission failure based on the received error messages. For example, after a network device receives a packet, it cannot access the network where the destination device resides, the network device automatically sends an ICMP Destination Unreachable message to the source.




```
[RTA]tracert 20.0.0.2

 traceroute to  20.0.0.2(20.0.0.2), max hops: 30 ,packet length:
40,press CTRL_C
 to break

 1 10.0.0.2        80 ms     10 ms     10 ms

 2 20.0.0.2        30 ms     30 ms     20 ms
```

Function: Tracert

Tracert checks the reachability of each hop on a forwarding path based on the TTL value carried in the packet header.

Tracert is an effective method to detect packet loss and delay on a network and helps administrators discover routing loops on the network.

Contents

Basic IP Address Configuration Commands

1. Enter the interface view.

[Huawei] **interface** *interface-type interface-number*

You can run this command to enter the view of a specified interface and configure attributes for the interface.

- *interface-type interface-number*: specifies the type and number of an interface. The interface type and number can be closely next to each other or separated by a space character.

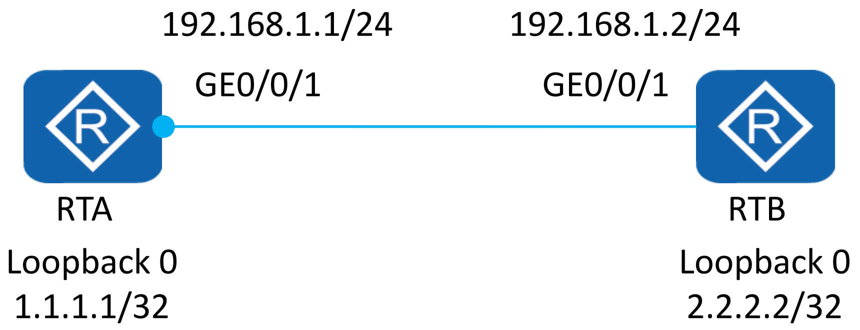
2. Configure an IP address for the interface.

[Huawei-GigabitEthernet0/0/1] **ip address** *ip-address* { *mask* | *mask-length* }

You can run this command in the interface view to assign an IP address to the interface on the network devices to implement network interconnection.

- *ip-address*: specifies the IP address of an interface. The value is in dotted decimal notation.
- *mask*: specifies a subnet mask. The value is in dotted decimal notation.
- *mask-length*: specifies a mask length. The value is an integer ranging from 0 to 32.

Case:
Configuring an IP address for an Interface



On the preceding network where the two routers are interconnected, configure IP addresses for the interconnected physical interfaces and logical IP addresses.

Configure an IP address for a physical interface.

[RTA] **interface** gigabitethernet 0/0/1

[RTA-GigabitEthernet0/0/1] **ip address** 192.168.1.1 255.255.255.0

Or,

[RTA-GigabitEthernet0/0/1] **ip address** 192.168.1.1 24

Configure an IP address for a logical interface.

[RTA] **interface** LoopBack 0

[RTA-LoopBack0] **ip address** 1.1.1.1 255.255.255.255

Or,

[RTA-LoopBack0] **ip address** 1.1.1.1 32

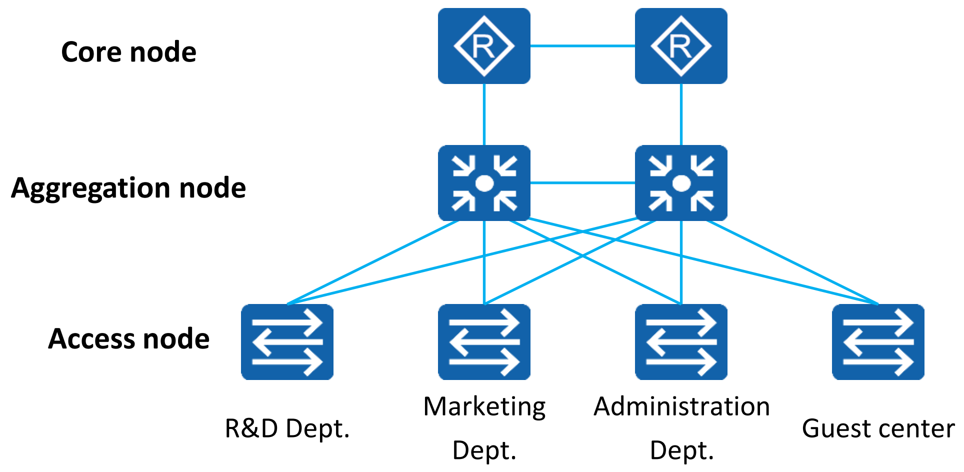
Network IP Address Planning

- IP address planning must be considered together with the network structure, routing protocols, traffic planning, and service rules. In addition, IP address planning should be corresponding to the network hierarchy and performed in a top-bottom way.
- In conclusion, IP address planning objectives are to achieve easy management, easy scalability, and high utilization.

Reference Planning Rules
Uniqueness, continuity, and scalability Structured and service-related

- **IP Address Planning Example**

Background	Address Type	Address Scope
Example: A company is assigned 192.168.0.0/16 as a network segment address.	Network segment of the R&D department	192.168.1.0/24
	Network segment of the marketing department	192.168.2.0/24
	Network segment of the administrative department	192.168.3.0/24
	Network segment of the guest center	192.168.4.0/24
	Others	...



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Quiz

1. (Multiple) Which class does 201.222.5.64 belong? (　　)

   A. Class A

   B. Class B

   C. Class C

   D. Class D

2. (Multiple) A company is assigned a class C network segment 192.168.20.0/24. One of its departments has 40 hosts. Which of the following subnets can be allocated? (　　)

   A. 192.168.20.64/26

   B. 192.168.20.64/27

   C. 192.168.20.128/26

   D. 192.168.20.190/26

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Summary

- To connect a PC to the Internet, apply an IP address from the Internet Service Provider (ISP).
- This presentation provides an overview of the IP protocol and describes concepts related to IPv4 addresses and subnetting.
- This presentation also describes the planning and basic configuration of IP addresses.

Thank You

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